3-W1　幂函数密度球：原点是否有奇异性

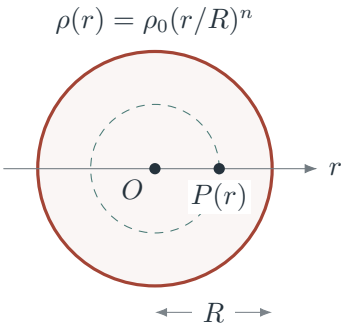
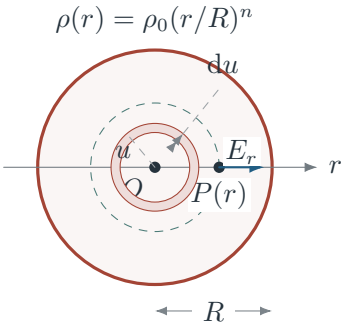


图 3-11　题图　幂律密度球与球形高斯面



$$Q(r)=4\pi\int_0^r \rho(u)u^2\,du$$

$$4\pi r^2 E_r = Q(r)/\varepsilon_0$$

先检验原点附近是否可积

图 3-12　解答分析图　逐壳累积电量，再用分段积分求势

3-W2　同心球壳：零场区与零电势面并不重合

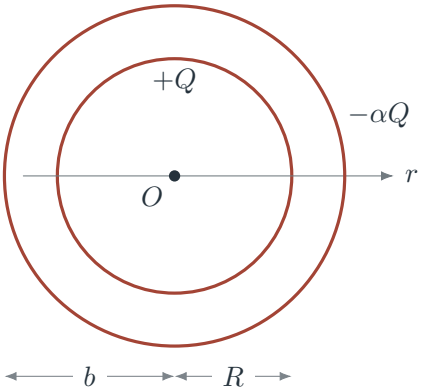
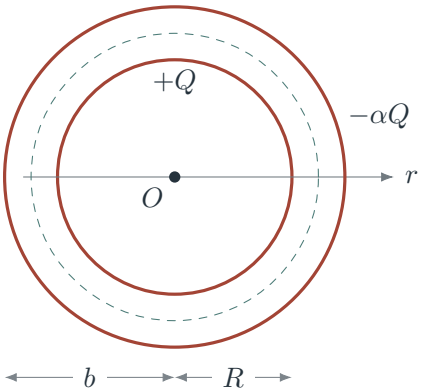


图 3-13　题图　带相反符号电量的同心球壳



$r < R:\ Q_{\text{in}} = 0$

$R < r < b:\ Q_{\text{in}} = Q$

$r > b:\ Q_{\text{in}} = (1-\alpha)Q$

两层的电势须相加

图 3-14　解答分析图　包围电量分三区读取，零势面另由电势确定

3-W3　非均匀圆柱：最大场不一定在表面

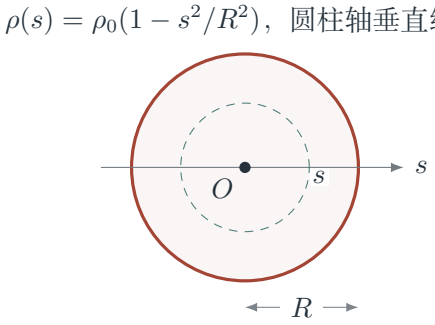


图 3-15　题图　非均匀无限圆柱的横截面

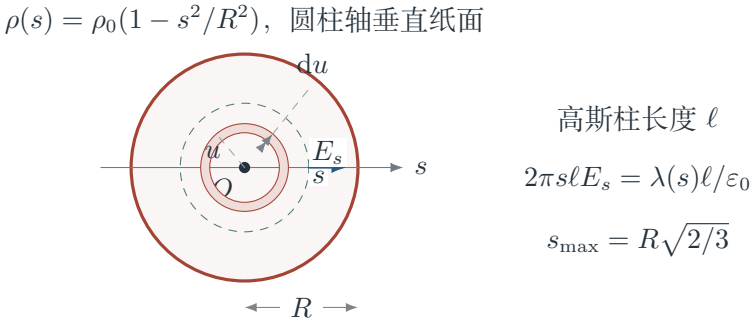


图 3-16　解答分析图　侧面通量决定场，内部极值由导数求出

3-W4　带电厚层与匀强外场的叠加

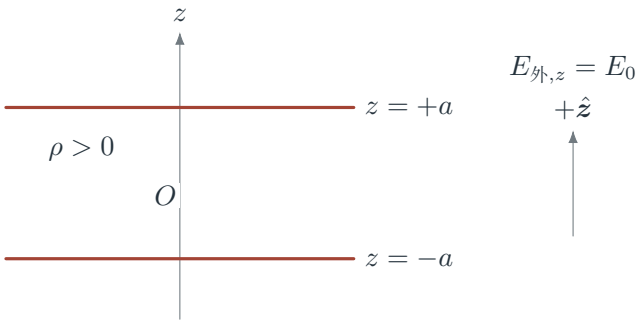


图 3-17　题图　均匀带电层叠加沿法线的外场

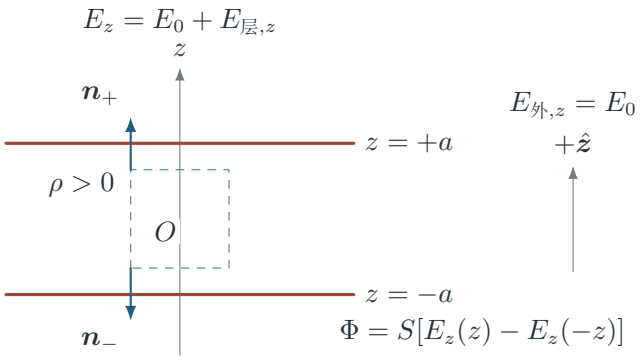


图 3-18　解答分析图　外场的上下通量抵消，场为零的位置会平移

3-W5　偏心球形空腔中的匀强场

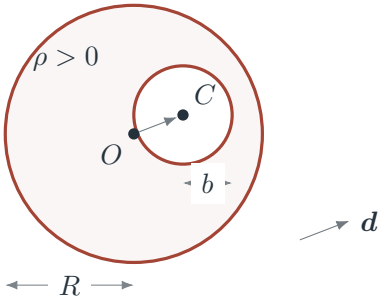


图 3-19　题图　均匀球中的偏心球孔

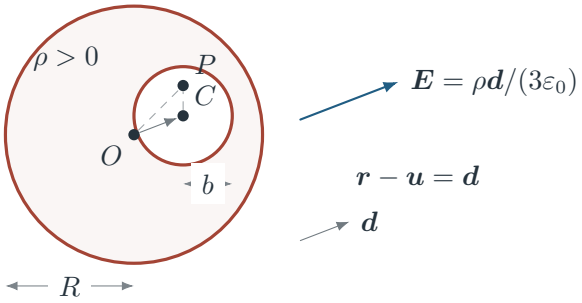


图 3-20　解答分析图　两个线性内场相减，孔内得到匀强场

3-W6　均匀带电圆柱中的偏心空腔

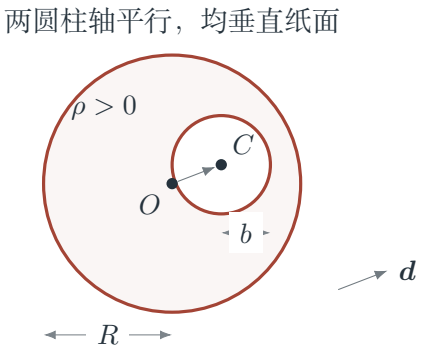


图 3-21　题图　无限圆柱中的平行偏心圆柱孔

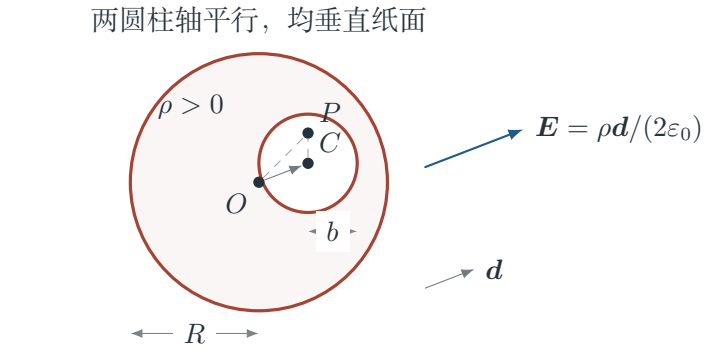


图 3-22　解答分析图　同样用叠加法，但内场系数是二分之一

3-W7　两块不等电荷薄板的分段场与势

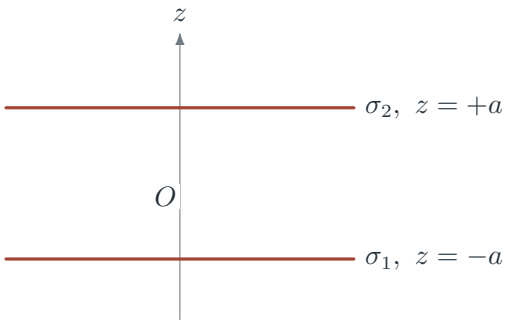


图 3-23　题图　两块任意面密度的无限平板

上板 σ_2，下板 σ_1；积分常数逐区匹配

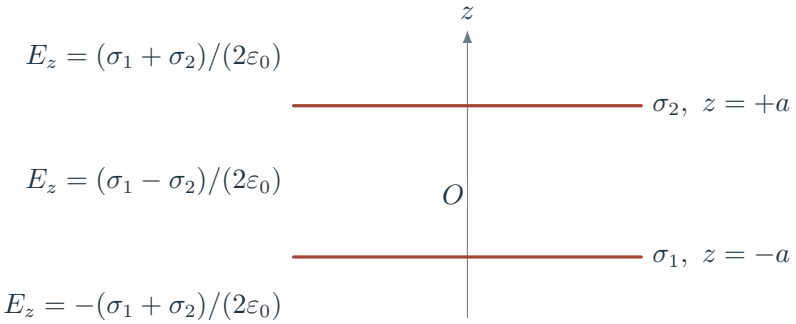


图 3-24　解答分析图　在各区对带符号的法向分量求和

3-W8　一个满足零净通量却未必是静电场的候选场

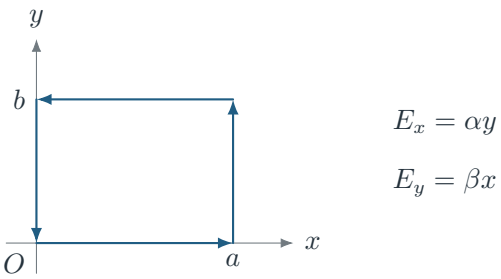
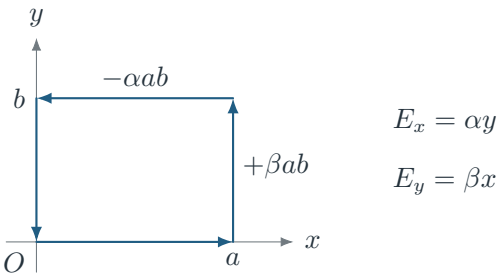


图 3-25　题图　候选场与逆时针矩形回路



$$\oint \boldsymbol{E} \cdot d\boldsymbol{l} = (\beta - \alpha)ab$$

图 3-26　解答分析图　分别计算四边，零散度还不足以判定静电场

3-W9　径向幂律场：保守性与平方反比的区别

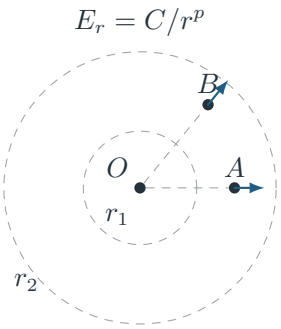


图 3-27　题图　仅在球形环域内讨论径向幂律场

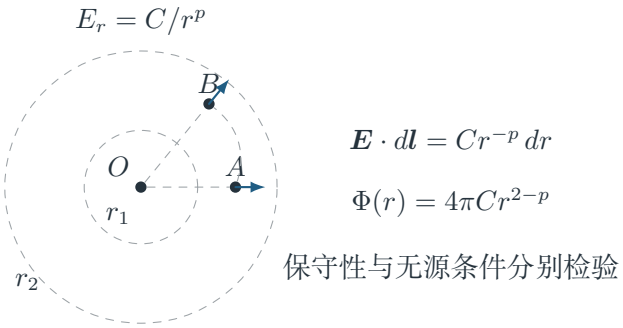


图 3-28　解答分析图　径向功只依赖端点，通量却随幂指数改变

3-W10　电势为零却仍有电场

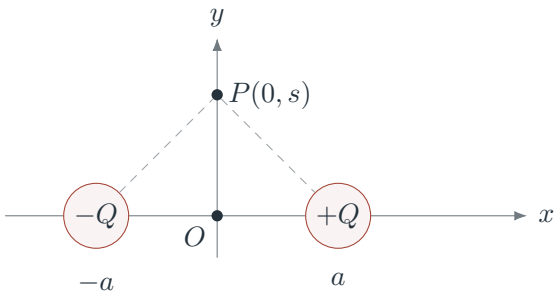
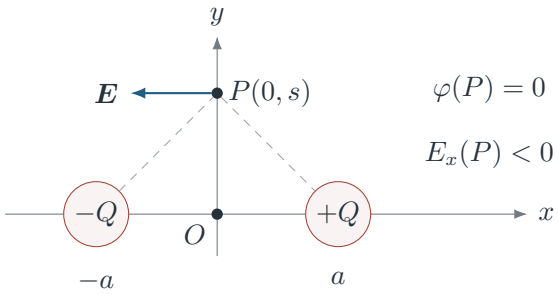


图 3-29　题图　异号等量点电荷的中垂面



等势面上可以存在垂直于面的场

图 3-30　解答分析图　电势抵消而水平场分量同向

3-W11　三千伏加速电子：符号、单位与近似精度

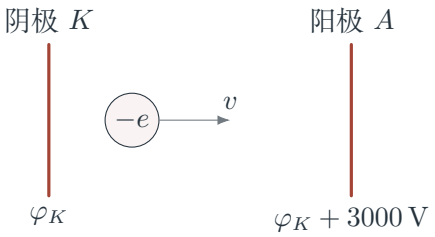
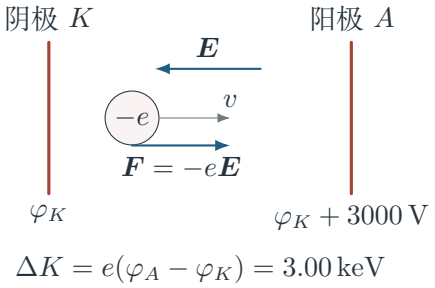


图 3-31　题图　从低势阴极出发的电子



$$\Delta K = e(\varphi_A - \varphi_K) = 3.00\,\mathrm{keV}$$

图 3-32　解答分析图　电子的力与场反向，动能随电势升高而增加

3-W12　均匀带电球内的三维恢复力

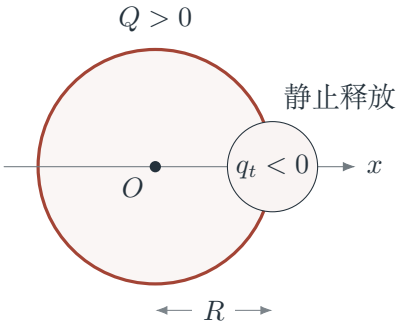


图 3-33　题图　可穿过固定正电球的负粒子

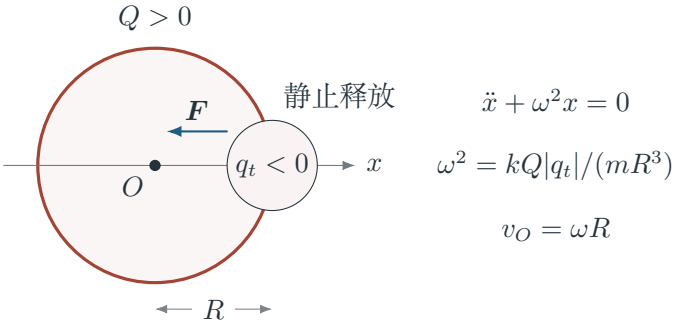


图 3-34　解答分析图　球内线性场产生三维简谐恢复力

3-T1　薄球壳的内场与常势

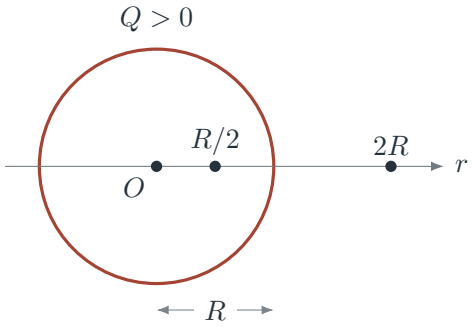
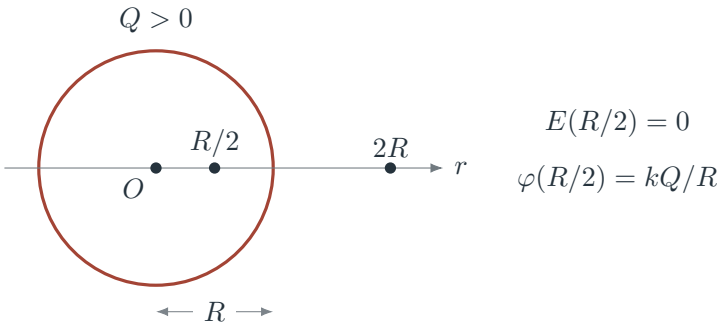


图 3-35　题图　薄球壳内外的两个观察点



$E_r(R^+)-E_r(R^-)=\sigma/\varepsilon_0$

图 3-53　解答分析图　内场为零仍可有非零常势，界面场有跃变

3-T2　均匀球中的势能变化

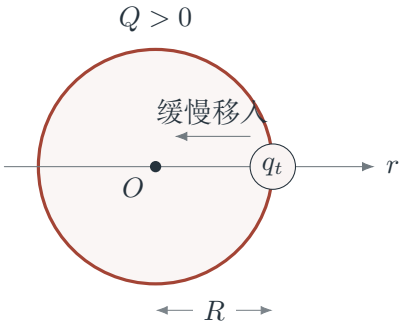


图 3-36　题图　试探电荷从表面缓慢移向中心

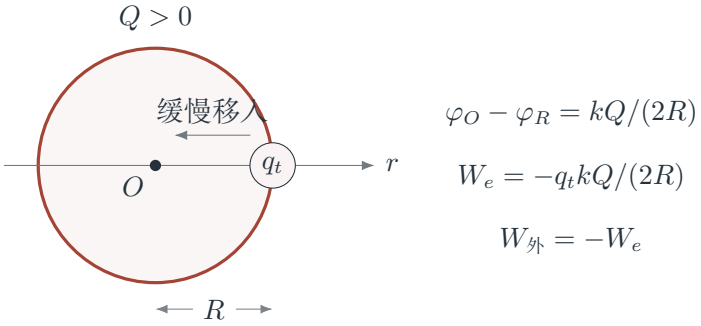


图 3-54　解答分析图　先求势差，再由电荷符号判定功

3-T3　密度与半径成正比

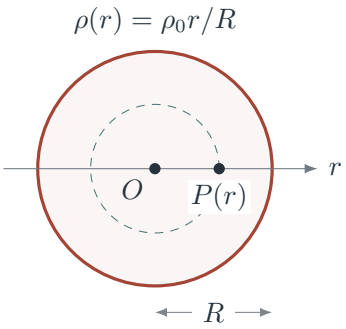
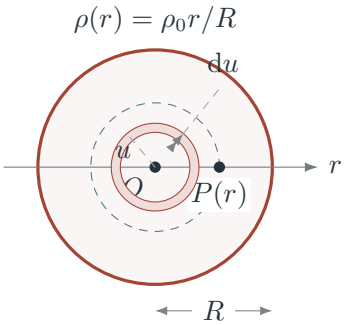


图 3-37　题图　密度从球心到表面线性增大



$$dQ = 4\pi\rho_0 u^3\,du/R$$

$$Q(r) = \pi\rho_0 r^4/R$$

局部密度不能代替体积平均密度

图 3-55　解答分析图　用球壳微元积分，内场正比于半径平方

3-T4　无限圆柱的有限参考

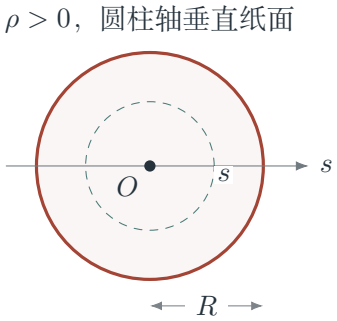
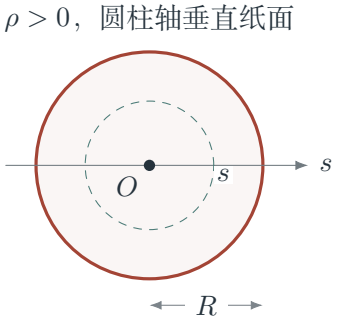


图 3-38　题图　均匀无限圆柱与同轴高斯柱



$$E_s=\rho s/(2\varepsilon_0),\quad s<R$$

$$E_s=\rho R^2/(2\varepsilon_0 s),\quad s>R$$

$$\Delta\varphi=-\int E_s\,ds$$

图 3-56　解答分析图　圆柱侧面面积与球面不同，势差用有限参考

3-T5　三块薄板的叠加

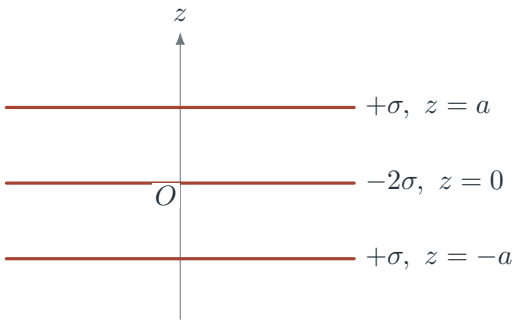


图 3-39　题图　总面电量为零的三板系统

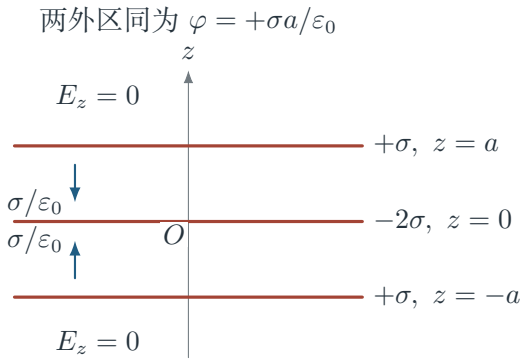


图 3-57　解答分析图　内区场方向相反，外区同势由积分得出

3-T6　检验候选界面场

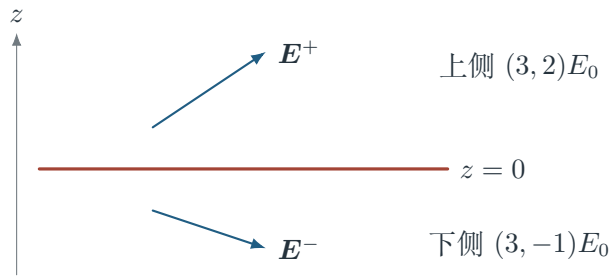
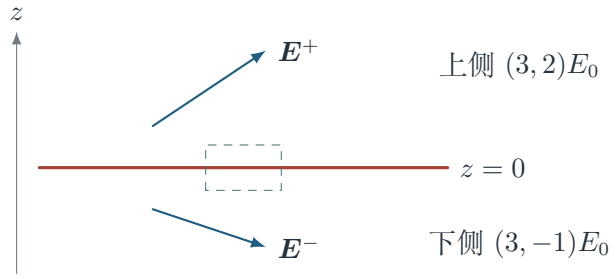


图 3-40　题图　普通面电荷两侧的候选场；分量顺序为 (x,z)

$$\sigma=\varepsilon_0(E_z^+-E_z^-)=3\varepsilon_0E_0$$



薄矩形回路要求 $E_x^+=E_x^-$

图 3-58　解答分析图　高斯盒检查法向跃变，回路检查切向连续性

3-T7　球孔两端的势差

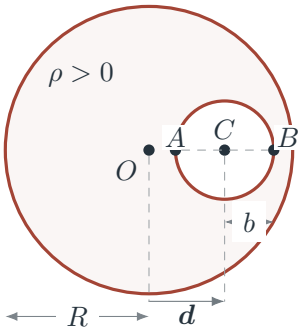
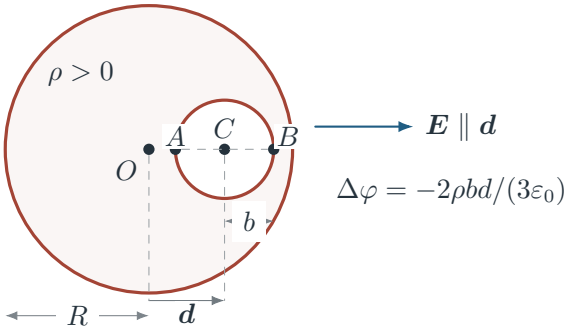


图 3-41　题图　孔内沿偏移方向的直径端点



沿场方向电势降低；功的符号还取决于 q_t

图 3-59　解答分析图　两端相距 $2b$，匀强场的功用位移点积

3-T8　两条异号线电荷的电势差

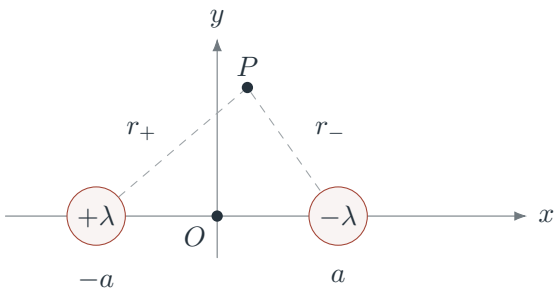


图 3-42　题图　两条异号无限线电荷的横截面

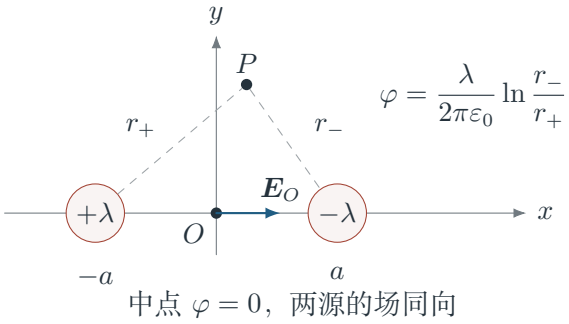


图 3-60　解答分析图　两项对数势合为距离比，符号按源电荷确定

3-T9　匀强场不改变点源净通量

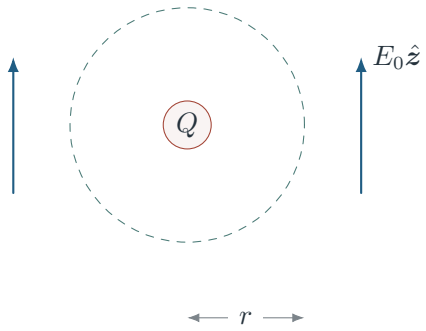


图 3-43　题图　点源周围的球面与叠加的匀强场

$\Phi = Q/\varepsilon_0$，局部总场并不均匀

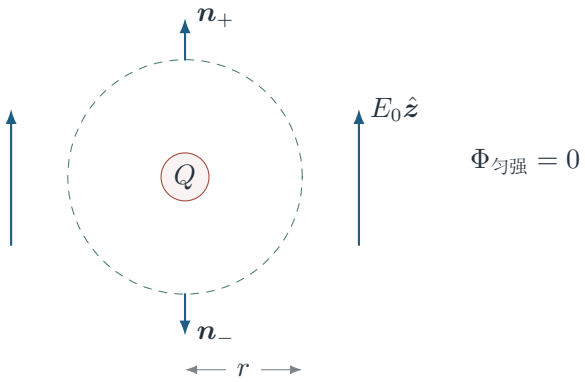


图 3-61　解答分析图　外加场的通量进出抵消，不能用净通量推总场大小

3-T10　用端点势求功

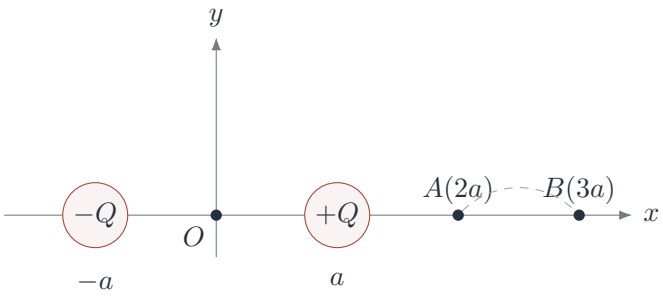


图 3-44　题图　异号点源外侧的两端点与任意路径

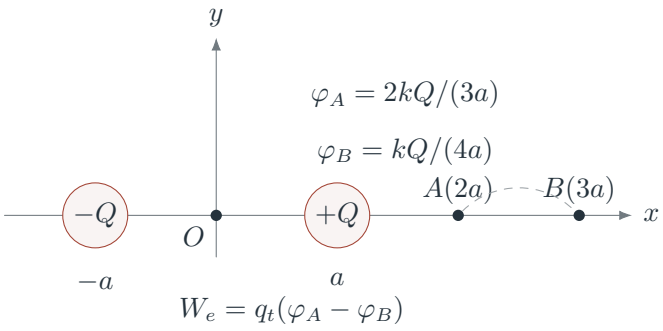


图 3-62　解答分析图　端点势差确定功，路径形状不进入结果

3-T11　两个线性分量构成的静电场

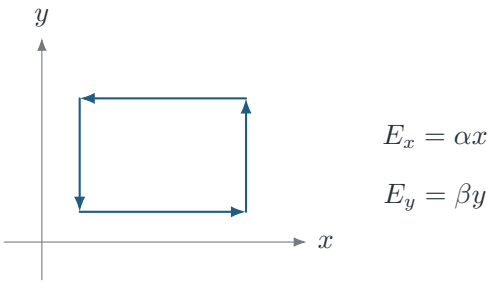
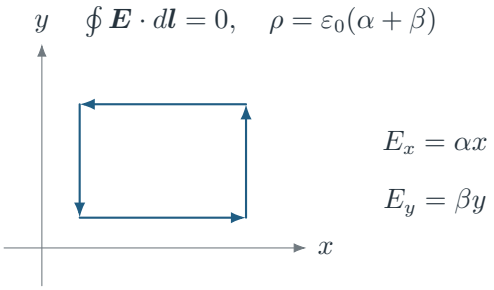


图 3-45　题图　两分量各自只依赖本方向坐标的候选场



$$\varphi = -\tfrac{1}{2}(\alpha x^2 + \beta y^2)$$

图 3-63　解答分析图　每对平行边的积分抵消，散度决定体电荷

3-T12　径向场与对数势

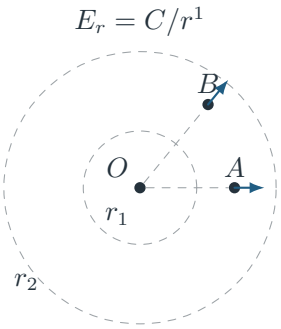


图 3-46　题图　随半径反比衰减的径向场

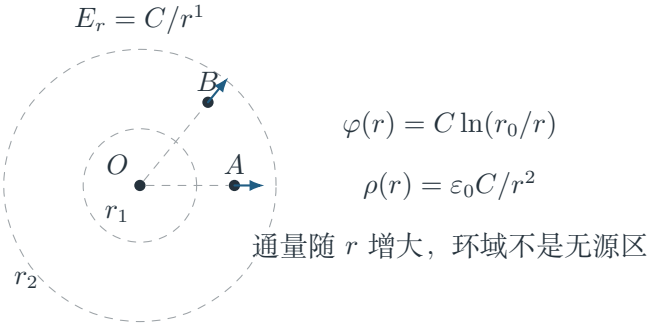


图 3-64　解答分析图　对数势描述功，变通量说明环域有体电荷

3-T13　挑战：电子的转折电势

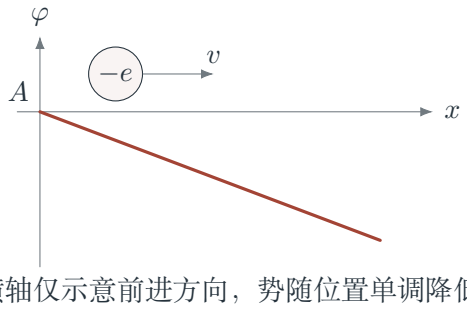


图 3-47　题图　电子进入电势逐渐降低的区域

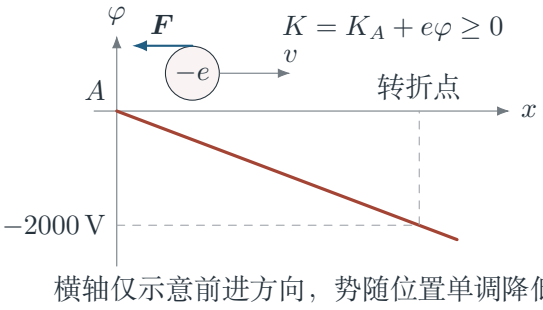


图 3-65　解答分析图　负电荷向低势运动时减速，转折点动能为零

3-T14　挑战：从分段电势反演电荷

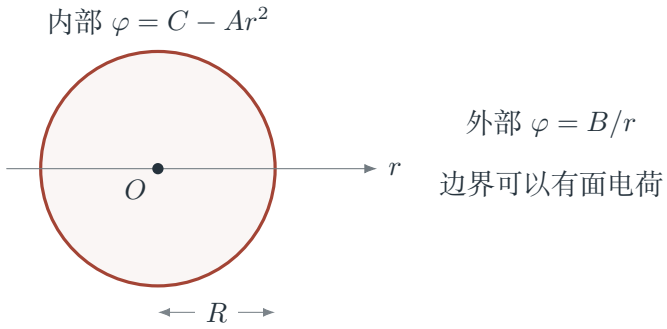


图 3-48　题图　给定球内外电势的反演问题

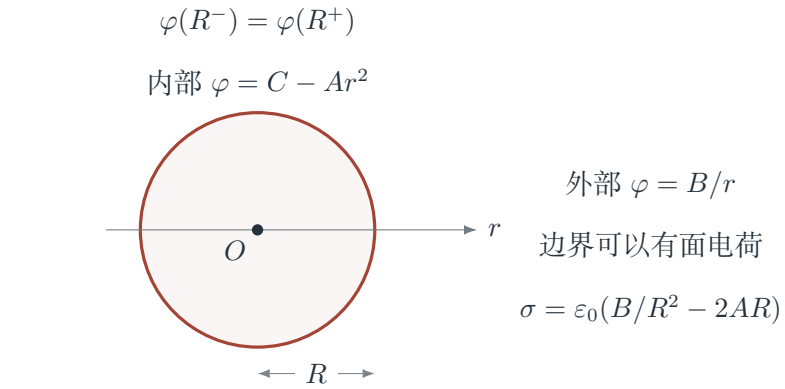


图 3-66　解答分析图　连续势与跃变场分别约束常数和面密度

3-T15　挑战：两个孔破坏了单孔匀强结论

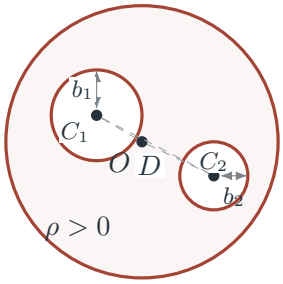
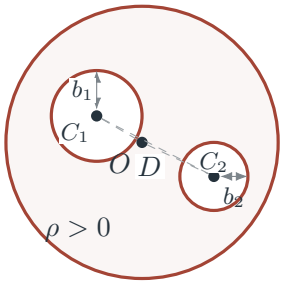


图 3-49　题图　互不相交的两个球孔



完整大球 + 负密度孔 1

+ 负密度孔 2

在 C_1：孔 1 自身场为零

孔 2 的外场随位置改变，孔 1 内一般不再匀强

图 3-67　解答分析图　完整球与两个负球叠加，不能直接套单孔匀强结论

3-T16　挑战：无源区内的三维静电稳定性

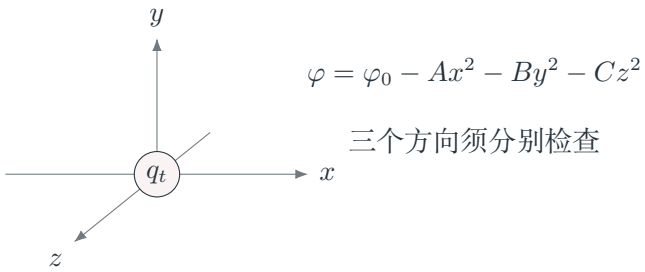
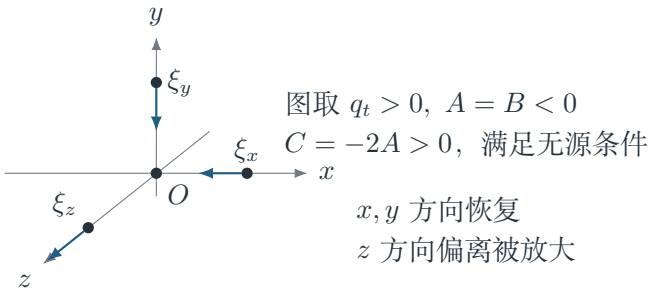


图 3-50　题图　平衡点附近的三个二次曲率



二次无源势的曲率和为零，不能在三个方向都恢复

图 3-68　解答分析图　取满足无源条件的一个示例，在三个微扰点画实际恢复或发散力。

3-T17　挑战：正负体密度组成的无限层

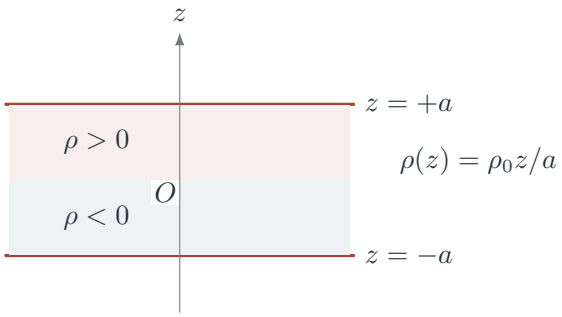


图 3-51　题图　上正下负的无限体电荷层

总面电量为零，内部仍有场与势差

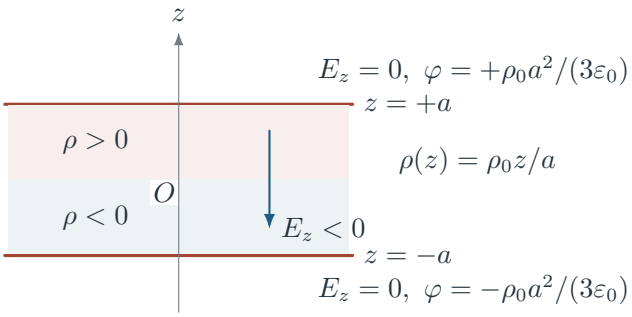


图 3-69　解答分析图　从远处零场积分进入层内，两个外区的常势不同

3-T18　挑战：从球外释放后的分段运动时间

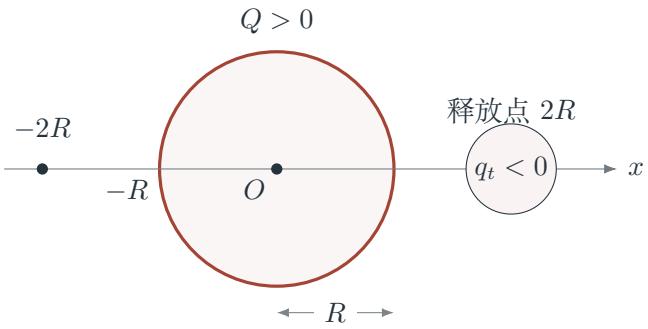
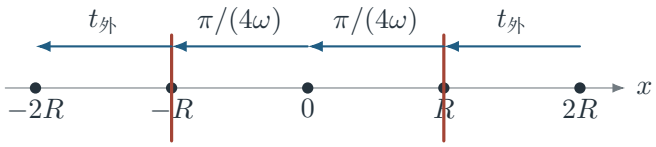


图 3-52　题图　从球外 $2R$ 静止释放，可穿过源的负粒子

右端到左端是半周期；返回后才是完整周期

到静止转折点　　从静止出发



$v=-\omega R$　　$v=-\omega R$

$$v=-\sqrt{2}\,\omega R$$

$$T=2(2t_{外}+t_{内}),\quad t_{内}=\pi/(2\omega)$$

图 3-70　解答分析图　按实际运动顺序分段计时；边界和中心速度均取向左的有向值。